

Algebraic Number Theory NMAL431

Exercise Sheet 1

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1. Let K be a field.
 - (a) Let $A \subset K$ be a discrete valuation ring with $\text{Frac}A = K$. Let π be a generator of its maximal ideal.
 - i. Show that every $x \in K^\times$ can be written uniquely as $x = u\pi^n$ with $u \in A^\times$ and $n \in \mathbb{Z}$.
 - ii. Define the function $v_A : K^\times \rightarrow \mathbb{Z}$ as $v_A(x) := n$. Show that v_A does not depend on the choice of π and that v_A is a valuation on K .
 - (b) Let v be a valuation on K , and let $A_v = \{x \in K^\times : v(x) \geq 0\} \cup \{0\}$. Show that A_v is a discrete valuation ring with $\text{Frac}A_v = K$.
 - (c) Prove that $A \mapsto v_A$ and $v \mapsto A_v$ are mutually inverse bijections between the subrings $A \subset K$ which are discrete valuation rings with $\text{Frac}A = K$, and the valuations on K .
2. This exercise is to classify all valuations on $\mathbb{Q}$.
 - (a) Let p be a prime, and define the function $v_p : \mathbb{Q}^\times \rightarrow \mathbb{Z}$ as $v_p(p^n r/s) := n$, where r, s are integers such that $\gcd(p, rs) = 1$. Show that v_p is well-defined and is a valuation.
 - (b) Let $v : \mathbb{Q}^\times \rightarrow \mathbb{Z}$ be a valuation. Show that $v = v_p$ for some prime p .
3. Let A be a (commutative, unital) ring and $S \subset A$ a multiplicative subset.
 - (a) Show that $S^{-1}A = 0$ if and only if $0 \in S$.
 - (b) Show that every subring of $\mathbb{Q}$ is of the form $S^{-1}\mathbb{Z}$ for some multiplicative subset $S \subset \mathbb{Z}$.
 - (c) Let A be a domain. Show that $A = \bigcap_{\mathfrak{m}} A_{\mathfrak{m}}$ inside $\text{Frac}A$, the intersection running over all maximal ideals $\mathfrak{m}$.
 - (d) Let $S = \{1, 2, 4, 8, \dots\} \subset \mathbb{Z}$. Show that $S^{-1}(\bigcap_n 2^n \mathbb{Z}) \neq \bigcap_n S^{-1}(2^n \mathbb{Z})$.
4. (The Chinese remainder theorem) Let R be a (commutative, unital) ring, and let $I_1, \dots, I_n \subset R$ be pairwise coprime ideals of R . Recall that two ideals $I, J \subset R$ are *coprime* if $I + J = R$.

- (a) Show that $\prod_i I_i = \bigcap_i I_i$.
- (b) Show that the natural map $R \rightarrow \prod_i R/I_i$ is surjective.
- (c) Deduce that there is an isomorphism $R/\prod_i I_i \cong \prod_i R/I_i$.
- (d) Let $m_1, \dots, m_n$ be pairwise distinct maximal ideals of R , and let $a_1, \dots, a_n \geq 1$ be integers. Show that the ideals $m_1^{a_1}, \dots, m_n^{a_n}$ are pairwise coprime.

5. Let R be a Dedekind domain.

- (a) Prove that R is a principal ideal domain if and only if it is a unique factorisation domain.
- (b) Suppose that R has only finitely many maximal ideals. Show that R is a principal ideal domain.
- (c) Show that any non-zero ideal of R is generated by at most two elements.

[*Hint*: Use Question 4]

6. Recall Gauss' lemma, which states the following: Let A be a principal ideal domain with field of fractions K and $f(X) \in A[X]$ a monic polynomial. If $f(X) = g_1(X)g_2(X)$ for some monic polynomials $g_1(X), g_2(X) \in K[X]$, then $g_1, g_2 \in A[X]$.

Prove that the same statement holds when A is assumed only to be a Dedekind domain. [*Hint*: Use localization and Question 3(c)]